Arjun Nanduri

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Education

University of California: Berkeley, BA in Computer Science — Expected May 2028

- GPA: 3.9 | SAT: 1590
- Courses: Probability & Stochastic Processes, Machine Learning, Computer Security, Data Structures

Research

Data Science Research Intern, CDSS, UC Berkeley – Berkeley, CA — Jan 2025 – May 2025

- Fine-tuned BART transformer for Ancient Akkadian lemmatization in low-resource settings, **beating open-source baselines by 15%** using a custom edit-distance metric over lemma forms.
- Built preprocessing and alignment pipelines for structured linguistic data from ORACC's cuneiform corpus ($\sim$**1M annotated words across** $\sim$**10K texts**).
- Stepped up to lead disorganized team with no formal authority — **built shared task system**, assigned work by strength, held mock presentations, presented at UC Berkeley Discovery Symposium.

Experience

Machine Learning Engineer, Aver Technologies Inc. – Virginia — Aug 2025 – Present

- Building a multi-stage PyTorch CV pipeline to automate depth tracking when driving piles (foundational support for heavy structures, like bridges) into the ground **— replacing a manual, hand-marked DOT compliance process**.
- Trained a CV pipeline for **real-time pile depth tracking** — object detection to localize the pile, YOLO segmentation and line detection to track painted depth markings, an open-source digit detector to identify line numbers, and audio-visual synchronization to isolate each hammer strike, converting pixel-space line displacement into inch/foot measurements via a depth map.
- Conducted field interviews with construction workers to surface real-world constraints — inconsistent hand-drawn markings, Spanish-speaking workforce — **directly informing model design decisions**.
- Field interviews also surfaced that a LIDAR-based measurement approach was cost-prohibitive given the project's thin margins, and that workers frequently crossed the camera's line of sight during recording **— informed the decision to pursue a CV-based approach over LIDAR** for cost-effectiveness.
- **Helped file a provisional patent**, designing a tripod-mounted 1000fps camera and Jetson Orin GPU capture rig.
- Designing two-phase roadmap: Phase 1 CV digitization of manual process, **Phase 2 LIDAR elimination of markings**.

Projects

Face Morpher — From-Scratch Warping (NumPy, dlib, scipy) — github.com/FavouritePlayer/face-morpher

- Implemented per-triangle affine solve (3-point correspondence $\rightarrow$ 2×3 matrix), inverse warping with bilinear interpolation, and piecewise Delaunay-triangle warping to produce morphs and mean faces; **all core geometry self-implemented in NumPy**, OpenCV limited to I/O and masking.

Context Custodian — Workspace Hygiene Agent (FastAPI) — github.com/FavouritePlayer/ContextCustodian

- Built a FastAPI agent that compacts and cleans an agent-first workspace for downstream agents to ingest, acting as the user via the Scalekit SDK; **every fix is a reversible move, never a delete**.
- Designed a three-stage prompt-injection detection pipeline — regex prefilter, vector-similarity narrowing, then LLM adjudication — with a **fail-open fallback to regex-only detection** if the LLM call fails, keeping live demos safe.
- Engineered honest-by-construction fix reporting: each fix call returns whether the action **actually executed live or only updated the local audit index**, so the UI never claims a success that didn't happen.

DealScout — Resale Arbitrage Agent (LangGraph, Playwright) — github.com/FavouritePlayer/dealscout-cloud

- Built a LangGraph-orchestrated agent that scores resale value with **deterministic arithmetic kept separate from LLM judgment** to avoid hallucinated pricing, then filters and ranks candidates through a structured-output LLM call.
- Restricted image-attachment to the post-filter candidate queue instead of the full scan, **avoiding a** $\sim$**4x blowup in scan time**.
- Diagnosed an anchoring bias in the valuation prompt — a shown reference price pulled the model's own estimate toward it — fixed by withholding it, **lifting genuine positive-detection rate** $\sim$**6x** (2/30 $\rightarrow$ 51/120).

Skills

Languages: Python, SQL

Technologies: PyTorch, Transformers (BART), NLTK, spaCy, NumPy, dlib, scipy, LangGraph, FastAPI, Playwright, Git